

PIE

A clean slice of exam security



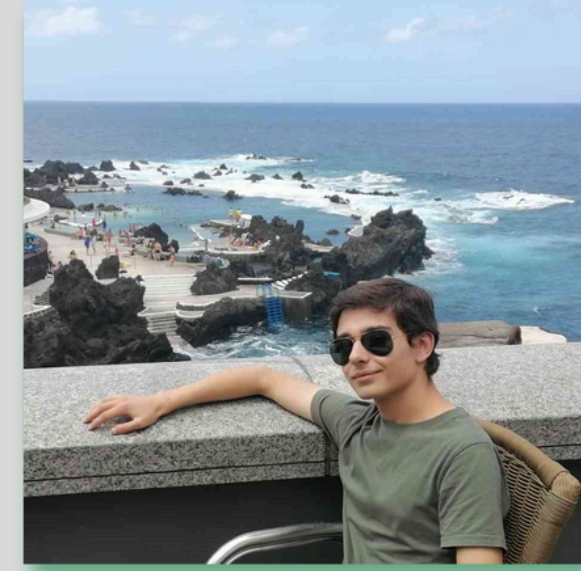
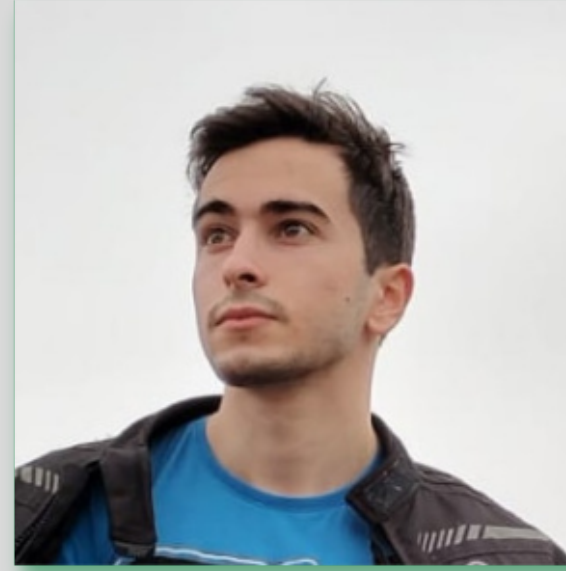
Portable Isolated Environment

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Who are we? Team



Who are we? Supervisors



Portable Isolated Environment

Motivation



Motivation

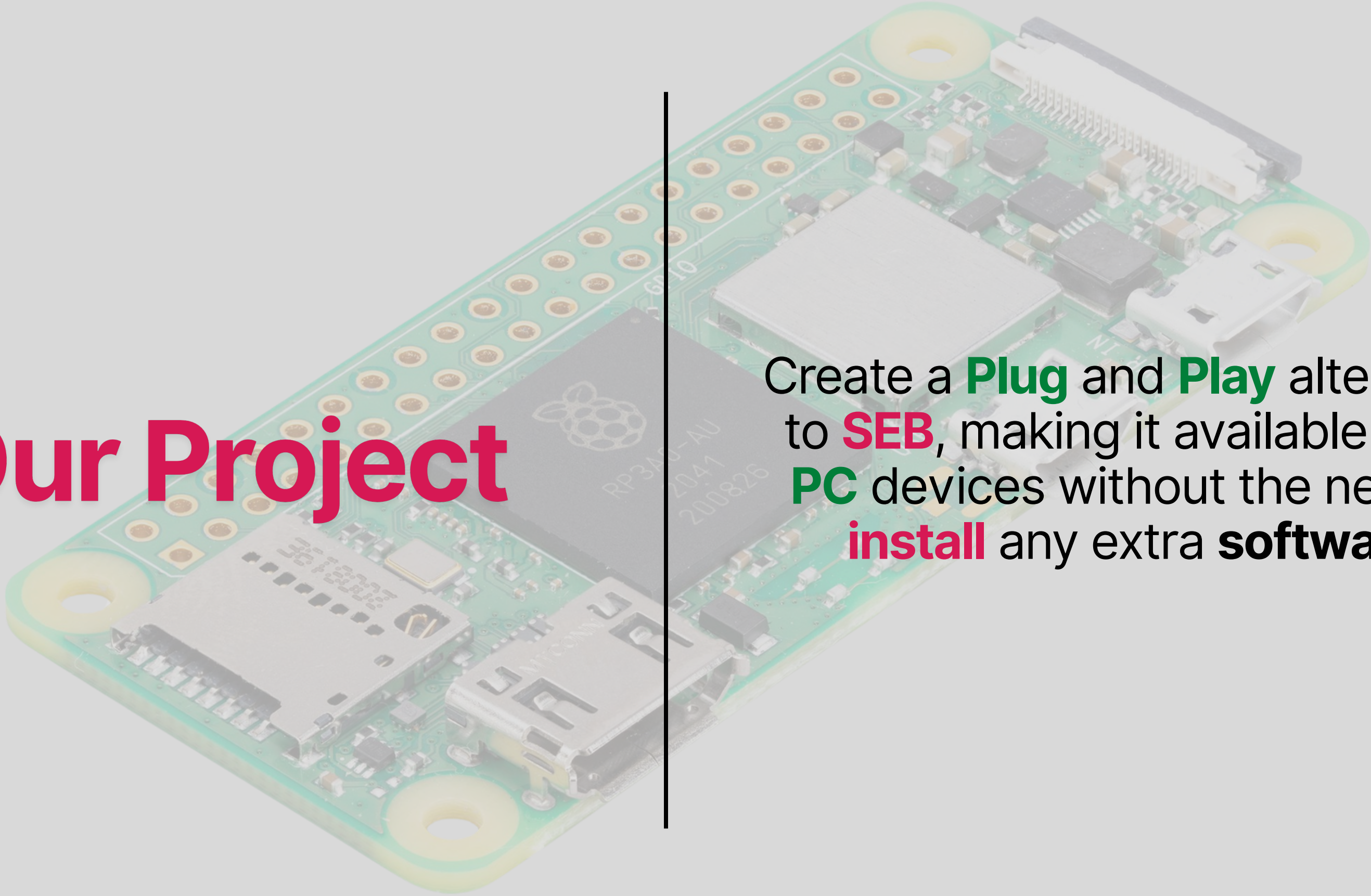
“**SEB** promised isolation. **Students** found the loopholes. It’s time for **PIE**.”

“When **students** know the **hacks** better than the **rules**, it’s time to change the system.”

“It’s not just outdated, it’s **outplayed** – **SEB** can’t keep up with today’s challenges.”

“**SEB** was built for yesterday’s exams. **PIE** is built for tomorrow’s **integrity**.”





Our Project

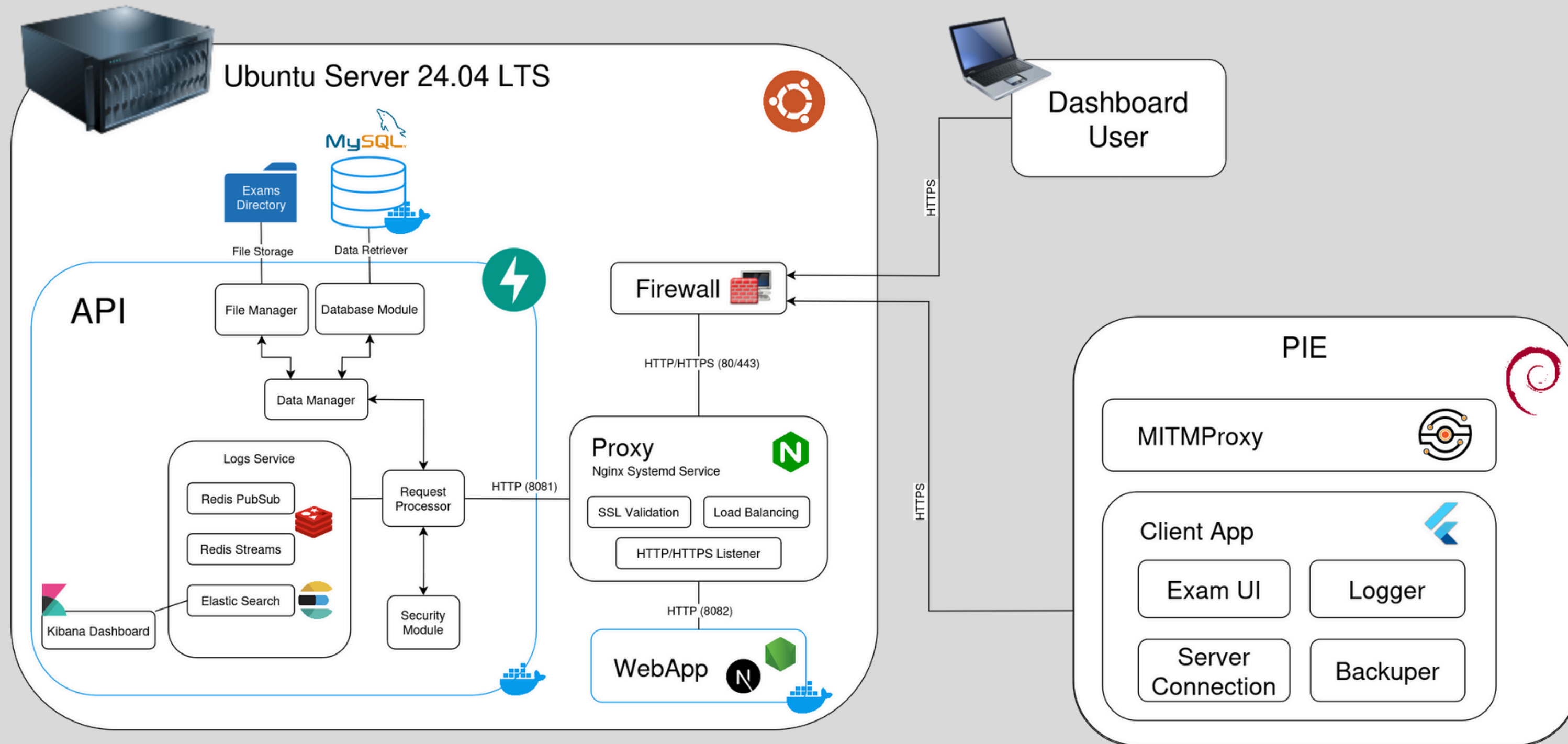
Create a **Plug and Play** alternative to **SEB**, making it available on **all PC** devices without the need to **install** any extra **software**.

Portable Isolated Environment

Architecture

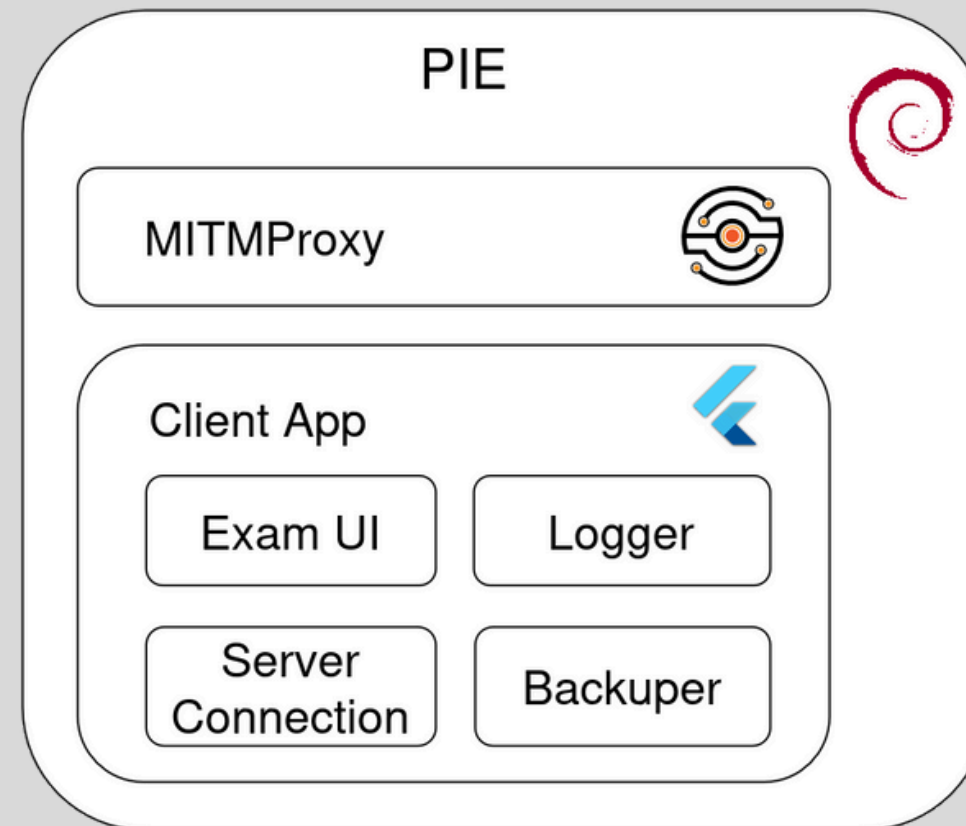


Architecture



Architecture **Student Side**

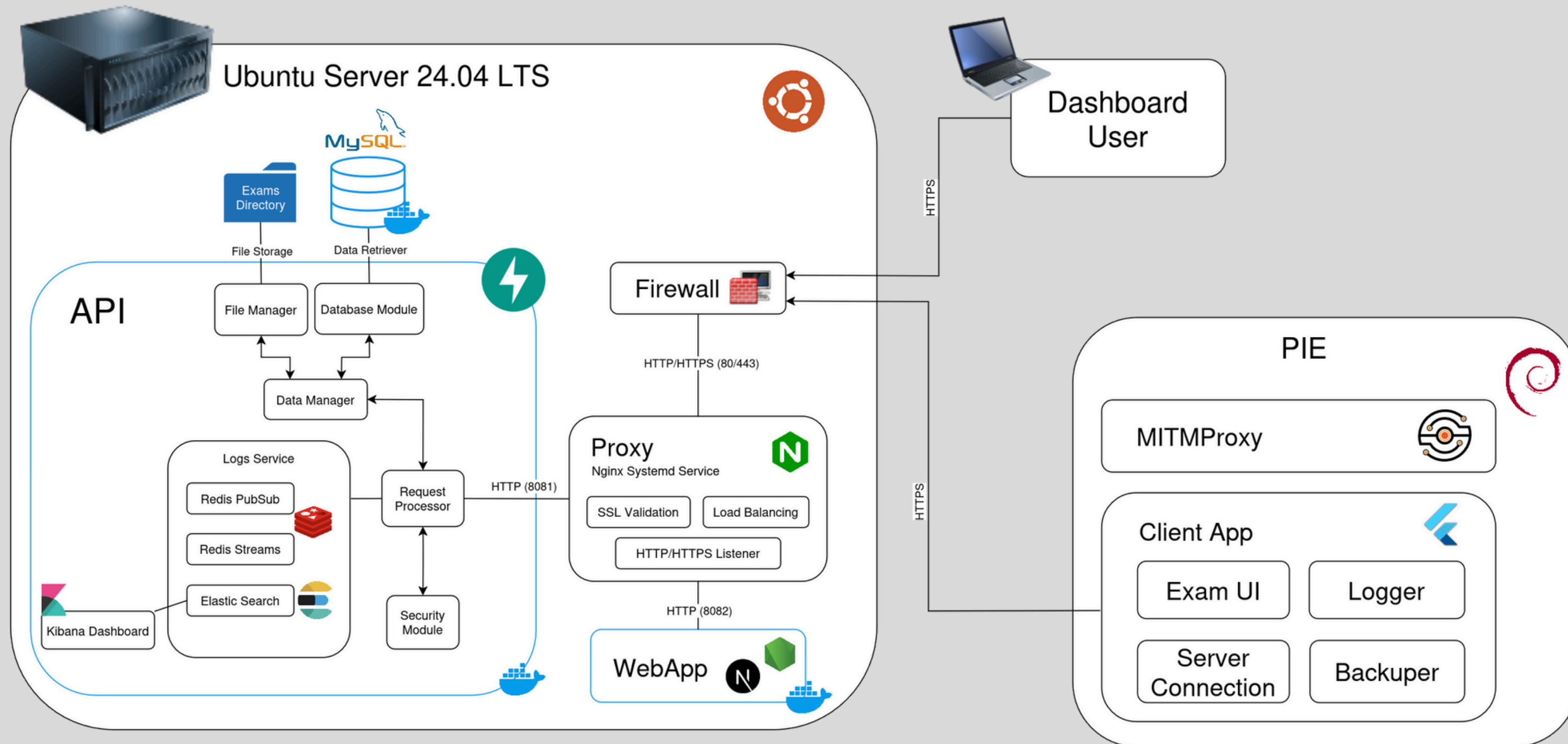
A secure **Debian** system on a **Raspberry Pi Zero W**, with **MITMProxy** managing network access.



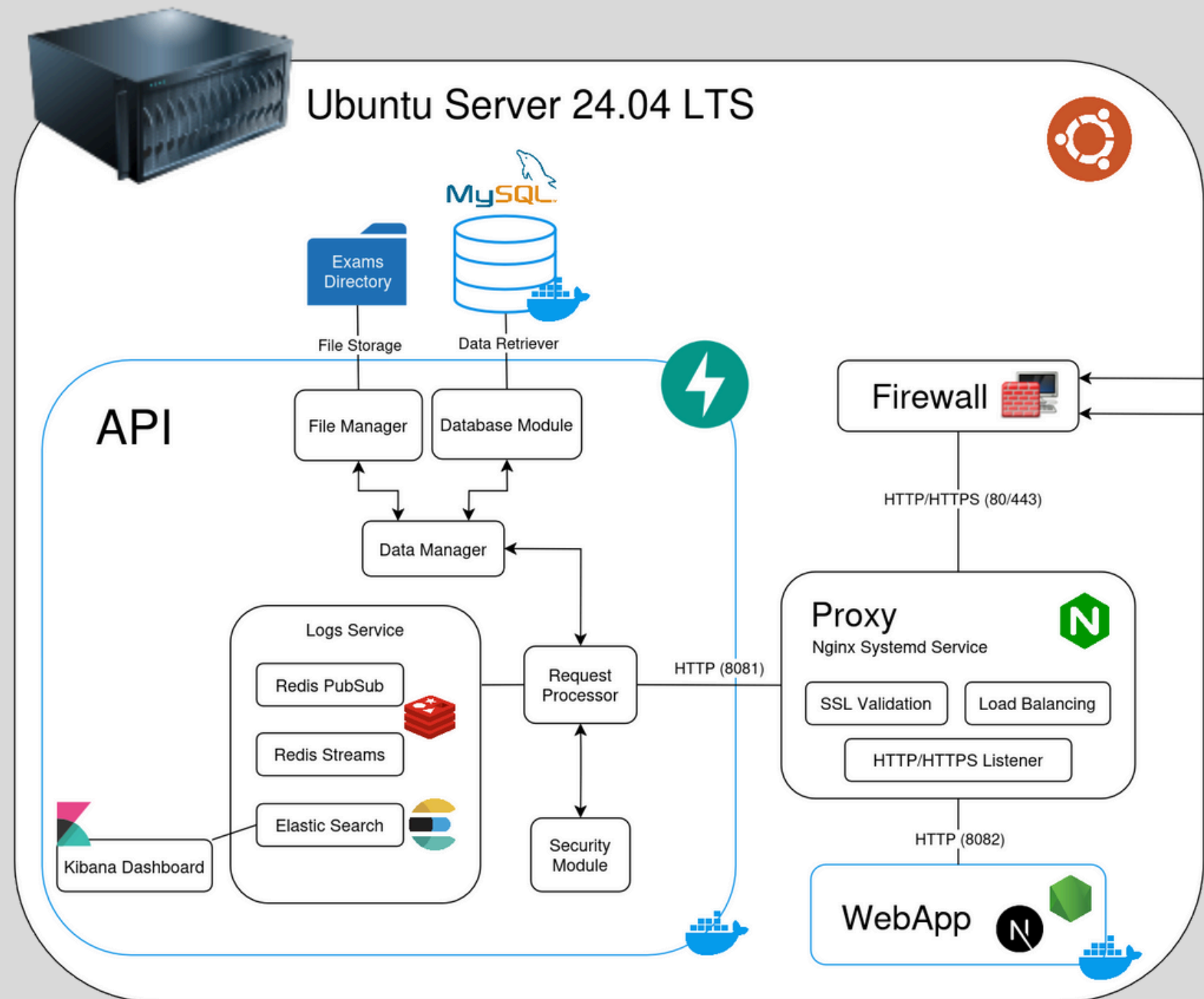
The **Client App** manages the entire **exam process**, providing the student interface, server communication, and ensuring everything runs smoothly.



Architecture



Architecture **Server Side**



The **API**, providing all the needed **services** and managing the stored **data**, and the **web application** behind a **safe proxy**, running on an **Ubuntu** machine.

Portable Isolated Environment

Security



Security **API**

User **Input Validation**

Files **Malware Analysis**

Exam Files **Encryption**

Secure Key Management

Filename **Sanitization**

CORS **policy Enforced**



Security **WebApp**

Auto **Log-off** for **inactivity**

Brute **Force Protection**

Protected Routes

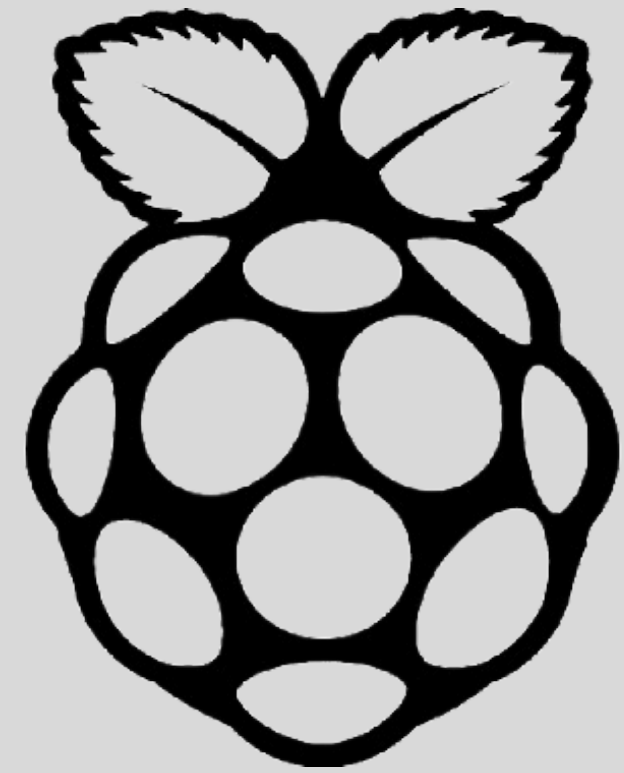
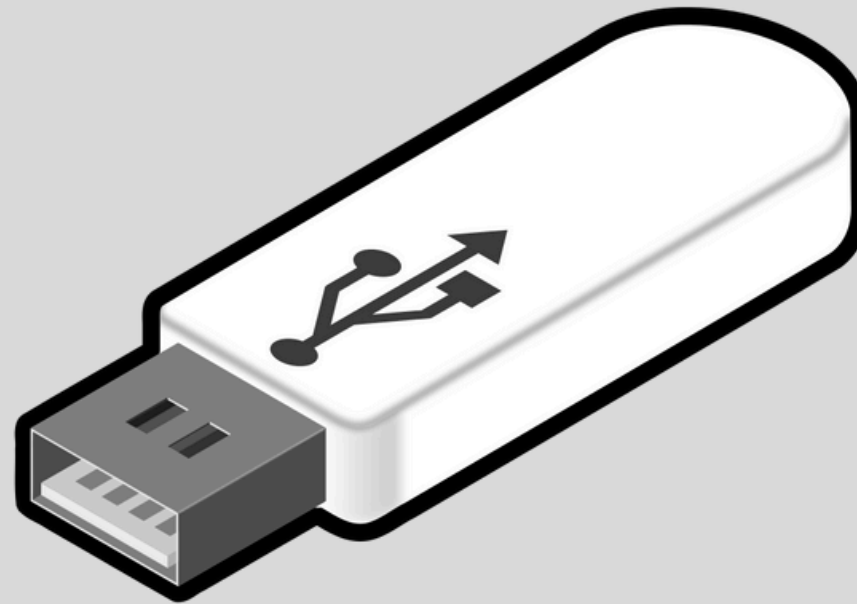
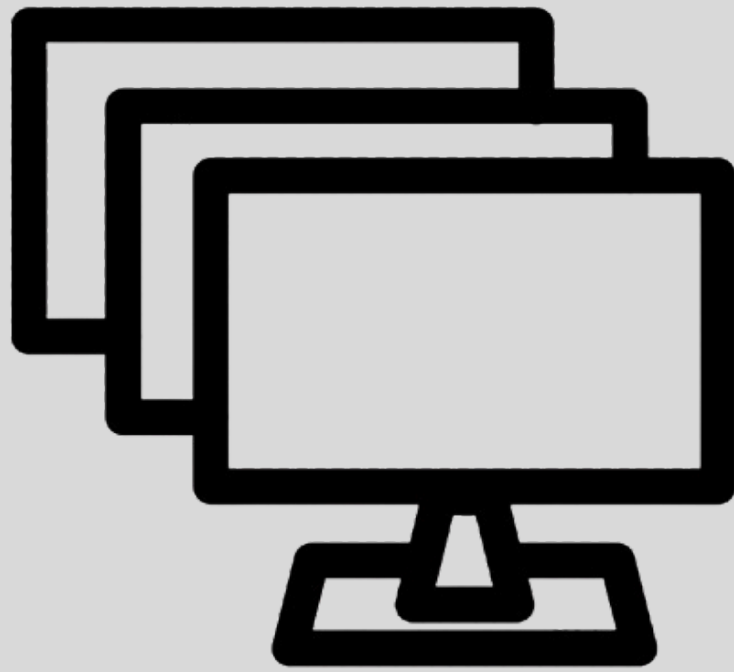
UA IdP **Authentication PoC**

HTTPS + **Secure Cookies**

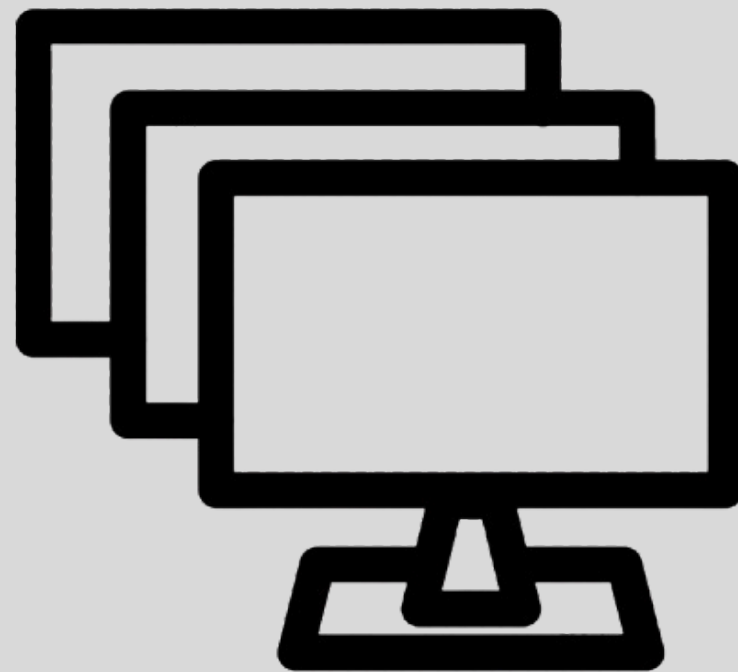
File Upload Security



Security Hardware

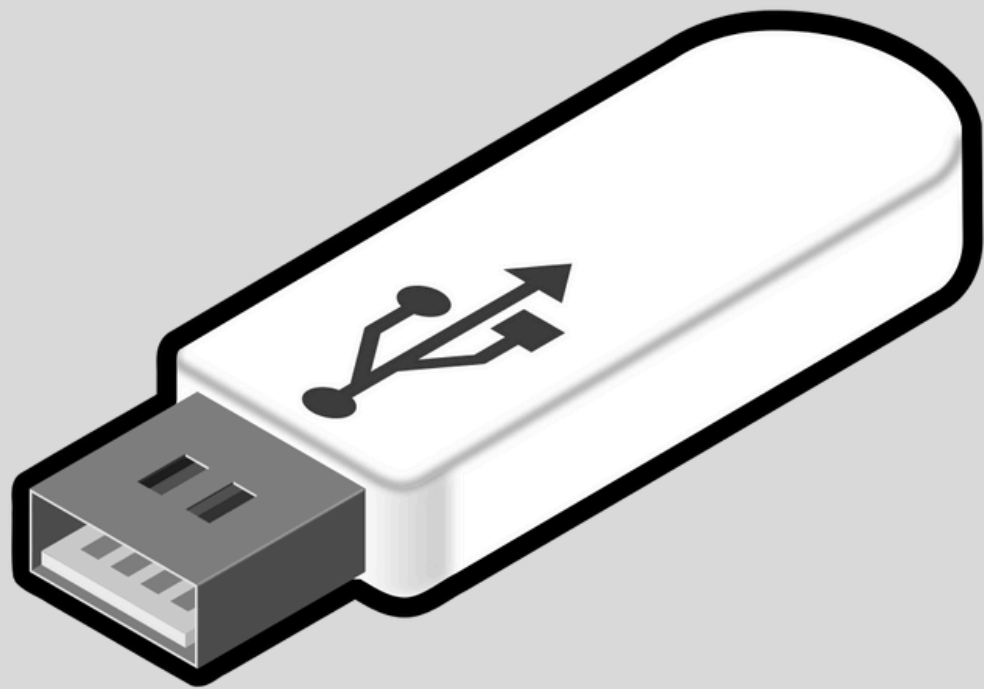


Security Hardware - V1



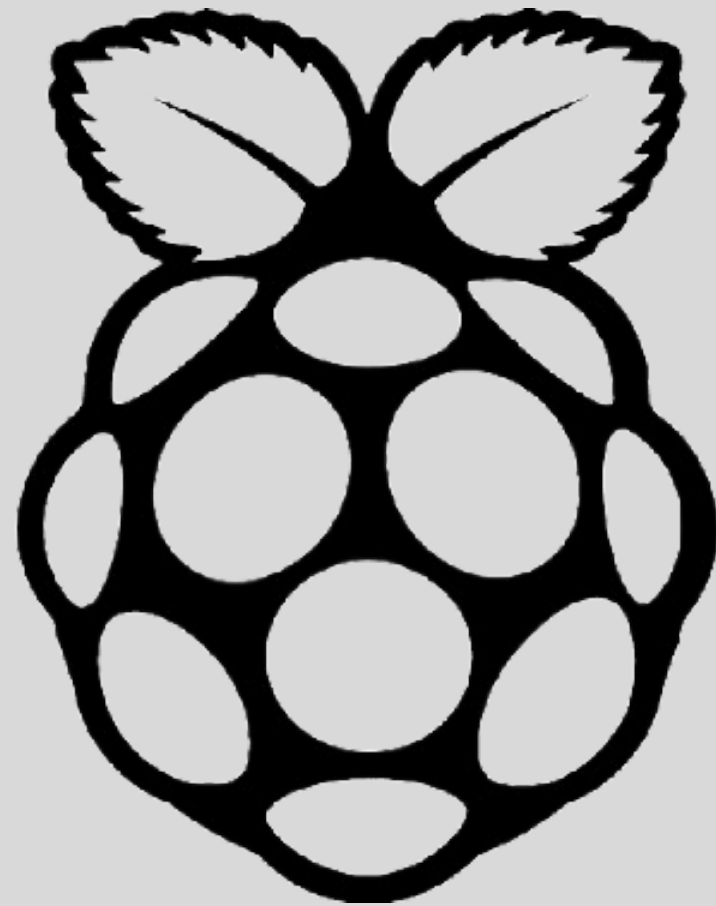
The **first steps** were to test the main ideas on a **VM**

Security Hardware - V2



Once the **image** had been created, using Live-Build, we moved on to **Live-Boot**.

Security Hardware - V3



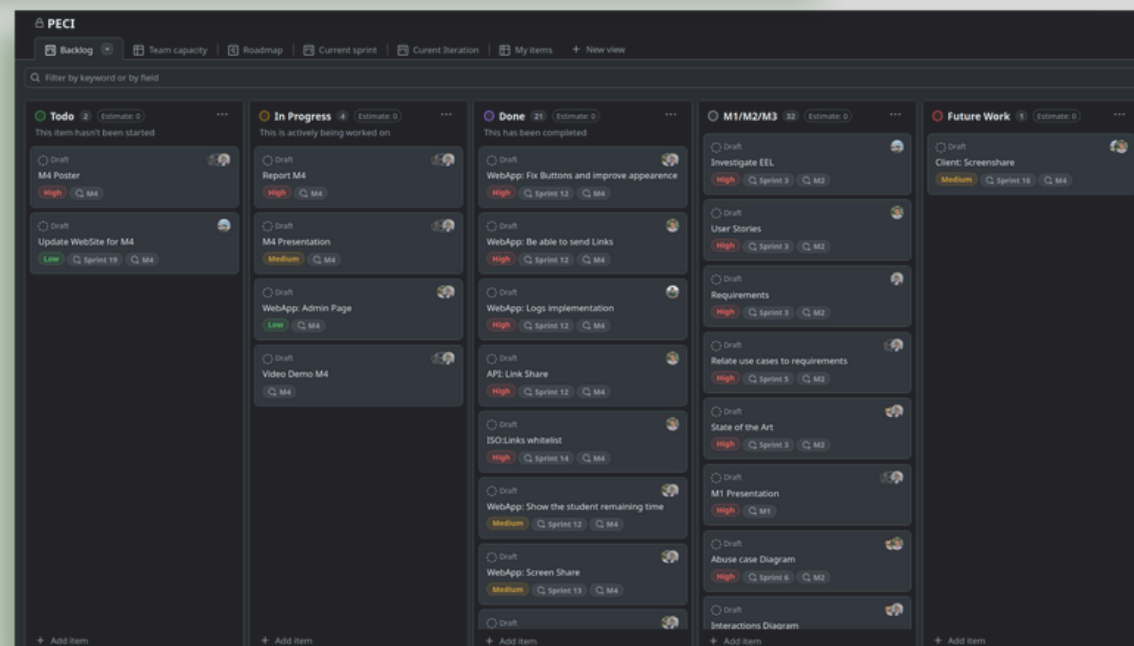
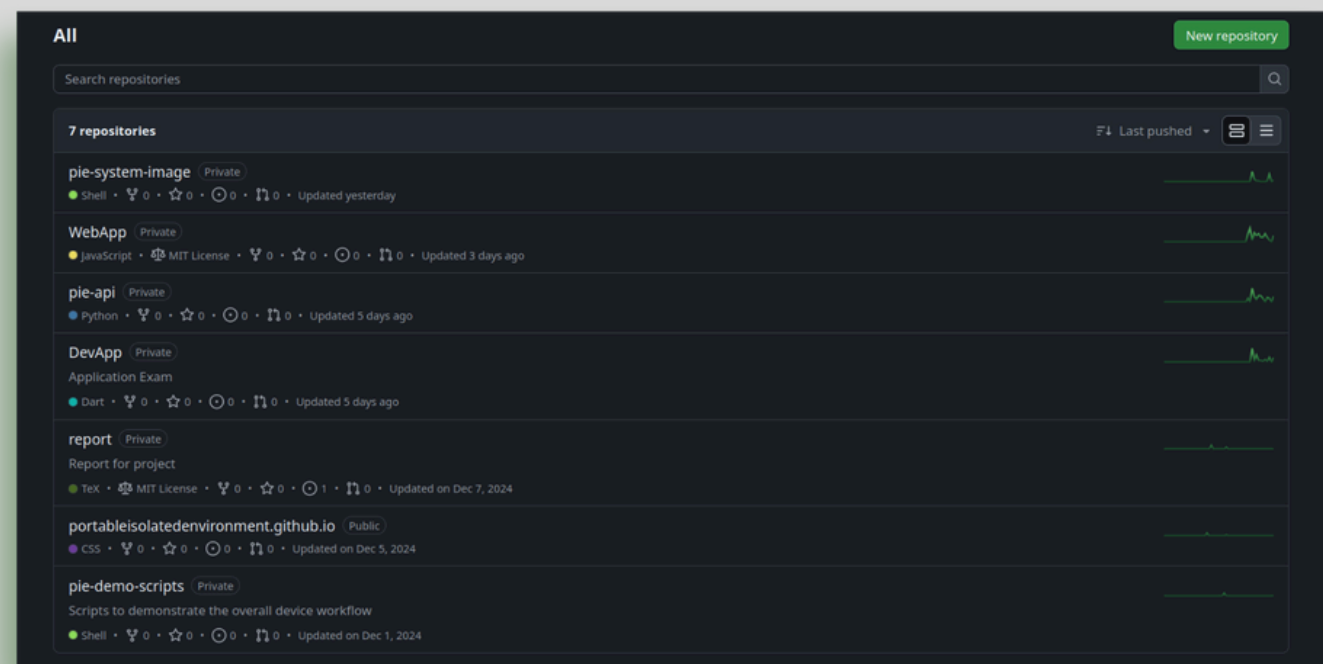
Finally, we set up the **raspberrys** with **gadget mode** to get to the final product

Portable Isolated Environment

BackLog



BackLog



More than
sixty issues,
seven repositories,
twenty-four branches and
thirty-thousand code lines
done in **one school year**



Future Work

Correct and **safe** implementation of students **screenshare** in **real-time** with teacher.

Optimizing the logs system, to be able to alert the teacher for **issues**.

Update both the **WebApp** and the **Client** server to be able to integrate any idp for **authentication**.

Overcoming the barrier of having to access the **BIOS** to boot from our system.

Remote centralized platform for **device** management.



Thanks! *Our page*

A clean slice of exam security



<https://portableisolatedenvironment.github.io/>